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Skimmed, Lactose-Free Milk Ingestion Postexercise: Rehydration Effectiveness and Gastrointestinal Disturbances Versus Water and a Sports Drink in Physically Active People

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Postexercise hydration is fundamental to replace fluid loss from sweat. This study evaluated rehydration and gastrointestinal (GI) symptoms for each of three beverages: water (W), sports drink (SD), and skimmed, lactose-free milk (SLM) after moderate-intensity cycling in the heat. Sixteen college students completed three exercise sessions each to lose ≈2% of their body mass. They drank 150% of body mass loss of the drink assigned in randomized order; net fluid balance, diuresis, and GI symptoms were measured and followed up for 3 hr after completion of fluid intake. SLM showed higher fluid retention (~69%) versus W (~40%; $p < .001$); SD (~56%) was not different from SLM or W ($p > .05$). Net fluid balance was higher for SLM (−0.26 kg) and SD (−0.42 kg) than W (−0.67 kg) after 3 hr ($p < .001$), resulting from a significantly lower diuresis with SLM. Reported GI disturbances were mild and showed no difference among drinks ($p > .05$) despite ingestion of W (1,992 ± 425 ml), SD (1,999 ± 429 ml), and SLM (1,993 ± 426 ml) in 90 min. In conclusion, SLM was more effective than W for postexercise rehydration, showing greater fluid retention for the 3-hr follow-up and presenting with low-intensity GI symptoms similar to those with W and SD. These results confirm that SLM is an effective option for hydration after exercise in the heat.

***Keywords**:* hydration, acute GI problems, fluid balance, dairy

Dehydration during prolonged exercise in the heat is a common occurrence requiring fluid and electrolyte ingestion to maintain homeostasis and prevent muscle fatigue, cramping, and heat exhaustion (Sawka et al., 2007). Postexercise rehydration involves ingesting solutions with water (W), carbohydrates (~60–80 g/L), sodium (~10–35 mmol/L), chloride (~10–12 mmol/L), potassium (~3–5 mmol/L), and an osmolality of ~280–380 mOsm/kg (Baker & Jeukendrup, 2011).

Gastric emptying of ingested beverages plays a key role in rehydration effectiveness (Leiper et al., 2001), representing a first barrier to the absorption process. Fluid volume, energy content, osmolality, and pH all have a direct impact on the speed of emptying of stomach contents. For instance, high-calorie beverages are emptied considerably slower in comparison with those with low—or null—energy content (Jeukendrup & Moseley, 2010). Intestinal absorption of fluids also influences rehydration speed; hydration beverages should aim to be emptied and absorbed quickly, together with having positive palatability, fluid retention, and gastrointestinal (GI) tolerance qualities (Baker & Jeukendrup, 2011; Pérez-Castillo et al., 2023).

Skimmed milk has been shown to be effective for hydration in an euhydrated state. Maughan et al. (2016) evaluated the hydration effectiveness of 13 beverages by calculating a beverage hydration index (BHI), using a reference value for W = 1.00. After correction for the W content of each beverage, oral rehydration solutions obtained a mean BHI of 1.50 ($p = .01$), while skimmed milk obtained 1.44 ($p < .01$), and whole milk 1.32 ($p = .02$), compared with W. This suggests that skimmed milk could be almost as effective as the solutions formulated specifically for hydration. Whole milk has resulted in better postexercise fluid retention than a sports drink (SD; Desbrow et al., 2014). Furthermore, skimmed milk has been shown to provide good rehydration after exercising in the heat, resulting in higher fluid retention than W and a SD (Shirreffs, Watson, et al., 2007).

Milk offers valuable nutrients for exercise recovery and/or hydration, like protein, sodium, potassium, calcium, and vitamins, but there is a concern with its lactose content due to GI issues. Lactose is a disaccharide that may facilitate the absorption of sodium and W, but it may cause severe problems in intolerant individuals (del Carmen Tocaa et al., 2022; Misselwitz et al., 2019). Even lactose-free skimmed milk may cause mild digestive issues like belching and bloating during exercise when 900 ml is ingested in 90 min (Aragón-Vargas et al., 2023). On the other hand, recent studies show that postexercise ingestion of dairy drinks is associated with effective rehydration and low severity of GI symptoms (Russo et al., 2021), although the ingested volumes have been moderate (≈1,700 ml over 4 hr). Other studies using higher volumes have not assessed GI symptoms (Volterman et al., 2014). These results warrant confirmation and the comparison of a large volume of lactose-free milk with W and conventional SDs. If a sufficiently large volume of milk causes greater GI distress than the latter two beverages, it will be difficult to recommend it as a rehydration beverage.

Therefore, the purpose of this study was to compare the rehydration effectiveness and GI responses of three widely

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available beverages: W, SD, and skimmed, lactose-free milk (SLM), ingested in a large volume after moderate- to high-intensity cycle ergometer exercise in the heat.

Methods

Sixteen college students, males and females aged 18–40 years old, were assessed as apparently healthy according to the Revised Physical Activity Readiness Questionnaire (Adams, 1999) and classified as physically active according to the American College of Sports Medicine criteria (Garber et al., 2011). Specific questions targeted possible cardiovascular, renal, or hepatic problems, as well as any pharmacological treatment which could influence the study results. Participants were requested to abstain from caffeine, alcohol, diuretics, stimulants, and strenuous exercise for 24 hr before each visit to the laboratory. Each participant provided his or her informed consent in writing; the protocol was approved by the University of Costa Rica Ethics and Science Committee, according to the form CEC-127-2022 and in compliance with the Declaration of Helsinki.

For each experimental session, participants arrived in the laboratory after fasting for 10 hr or more, on three different occasions separated by at least 48 hr. They recorded their dietary intake before the first session with the intention of replicating it for the remaining visits. They also followed specific fluid intake instructions for the preceding 24 hr. A urine sample was obtained upon arrival to test for urine specific gravity with an ATAGO URC-Ne refractometer (d 1.000–1.050), accepting values ≤1.020 as euhydration. Each participant consumed a standard breakfast providing 1,573 kJ (376 kcal): 11% fat, 14.5% protein, and 74% carbohydrate, including 200 ml of fluid and ~876 mg sodium. After 30 min, they were weighed nude and dry (preexercise body mass, BM_{pre}) on an e-Accura DSB291 scale to the nearest 10 g. At this point, all baseline GI symptoms were evaluated with the questionnaire used by Aragón-Vargas et al. (2023) on a Likert scale going from 0 (*no problem*) to 9 (*the worst it has ever been*).

Exercise sessions consisted of intermittent (20 min) pedaling on a stationary bicycle (Schwinn AC Performance Plus) in a heat chamber (32 °C, 70% relative humidity) at a moderate to high intensity (80%–85% maximum heart rate); these 20-min exercise bouts were repeated until BM loss reached ~2% BM_{pre}. Exercise intensity was controlled via heart rate monitoring with a Polar (Model FT7). Upon exercise completion, participants were weighed nude and dry again ($BM_{postexer}$) and the GI problems questionnaire was applied again. Conditions were assigned in random order: bottled W, SD (Gatorade, 60 g carbohydrate/L, 18 mEq Na^+/L, 3 mEq K^+/L, 1,000 kJ/L, specific gravity [SG] = 1.028), or SLM (Delactomy, 48 g carbohydrate/L, 2 g fat/L, 32 g protein/L, 17 mEq Na^+/L, 40 mEq K^+/L 1,415 kJ/L, SG = 1.036). Fluid volume for ingestion was calculated as 150% BM loss and divided into three aliquots to be ingested every 30 min, weighed with an OHAUS compact digital balance with 1 g accuracy (model CS2000). At the end of this 90-min rehydration period, a third nude and dry BM was obtained ($BM_{postingestion}$). All participants were evaluated under all conditions, with a minimum of 48 hr between tests for the same participant.

After rehydration, urine was collected for 3 hr at 0, 30, 60, 90, 120, 150, and 180 min. Each sample was kept in a preweighed container, with the gross weight recorded using the OHAUS precision scale mentioned above. Net weight was obtained by subtracting the container, and volume was calculated in milliliters assuming 1 g = 1 ml (Kurdak et al., 2010). At 3 hr, nude BM was recorded (BM_{3hPI}) using an e-Accura scale (Model DSB921, 250 kg, 10-g resolution). Total urine volume (TUV) was calculated by summing the seven samples. From TUV and ingested volume, percent fluid elimination (%FE) and conservation (%FC) were calculated, according to:

- %FE: (TUV × 100)/fluid ingestion volume
- %FC: 100%−%FE

Net fluid balance (NFB) was calculated four times: preexercise, postexercise, postingestion, and 3 hr postingestion, using BM immediately prior to exercise, as follows:

$$NFB_{time} = BM_{time} - BM_{pre}.$$

Finally, at times 60, 120, and 180 min postexercise, the GI symptoms questionnaire was applied, for a total of five GI symptoms evaluations. Drawing on Aragón-Vargas et al. (2023) and Pfeiffer et al. (2012), this questionnaire assesses GI symptoms across three categories: upper tract (reflux, bloating, nausea, etc.), lower tract (cramps, flatulence, etc.), and systemic (dizziness, headache, etc.). In addition, a combined score (CS) adding up all the values for each symptom under each of the three categories (maximum potential score of 63 for upper GI, 54 for lower GI, and 36 for systemic symptoms) was calculated. These were standardized using the formula:

$$\text{Standardized } (z) \text{ score}: zCS = \frac{CS - \text{Mean CS category}}{\text{Standard Deviation CS category}}.$$

Statistical Analysis

This was an experimental, repeated-measures, randomized (conditions), and crossover study. Descriptive statistics (mean, *SD*, and range) were calculated for age, height, initial BM, and GI symptoms to characterize the sample; variables were evaluated for normality (Shapiro–Wilk test). One-way analyses of variance on urine gravity, initial mass, fluid volume, exercise time, environmental conditions (temperature and humidity), and GI symptoms (pre- and postexercise) verified baseline similarity among conditions; Levene's test verified homoscedasticity. Dependent variables: TUV, percent dehydration, and %FC were each analyzed with a one-way analysis of variance. For the following variables that had to be analyzed for more than one independent variable, least squares were applied: NFB was compared with double interaction (Condition × Time of Measurement) and with participants as a random effect. Both partial and accumulated urine volumes were examined by weighted least squares with a double interaction (Condition × Time of Measurement) and participants as a random effect; for the purpose of analysis, urine samples were collected every 30 min for 3 hr, resulting in seven collection time points. For statistical analysis and to avoid too many zero values, these points were grouped into four periods: postingestion; 1, 2, and 3 hr postingestion. Last, GI symptoms were evaluated with the standardized CS (*z*PC) for each category, using least squares, a triple interaction (Condition × Time of Measurement × Symptom) and the participants as a random effect. Analyses were performed with RStudio 2023.06.0 + 421 (figures) and JMP Pro 17 (SAS Institute, Inc.).

Results

Sixteen participants (four females and 12 males) completed all three conditions in this study. Their basic characteristics were (mean ± *SD*; min–max): age = 22.9 ± 2.5; 19–28 years. Height = 168.9 ± 8.9; 150.2–182.6 cm. Weight = 66.6 ± 13.9; 49.9–107.1 kg. A post hoc

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statistical power (sp) analysis confirmed that a sample of 16 participants, with sp > 0.85 to be considered adequate under $\alpha < .05$, resulted in the following sp values considering specific meaningful mean differences (mmd) for each variable: TUV, mmd = 300 ml, sp = 0.994; fluid conservation, mmd = 10%, sp = 0.87; NFB, mmd = 0.2 kg, sp = 0.94; and GI symptoms (upper, lower, and systemic), mmd = 0.6 arbitrary units (AU), sp > 0.90.

One-way analysis of variance showed that W, SD, and SLM trials were performed under similar hydration status (Table 1): Baseline urine specific gravity ($p = .581$) and BM ($p = .998$) were the same. During exercise, both ambient heat ($p = .529$) and relative humidity ($p = .708$) were similar among conditions; resulting percent dehydration ($p = .715$) and exercise time ($p = .999$) were not different among conditions. GI symptoms (upper, lower, and systemic) showed no significant differences, either pre- or postexercise, among the experimental conditions. Levene's test confirmed homoscedasticity within variables ($p > .05$).

Hydration Effectiveness

Figure 1 shows the changes in NFB. The exercise protocol resulted in similar losses of ~2% BM: $NFB_{postexer}$ = −1.29, −1.29, and −1.31 kg prior to conditions W, SD, and SLM, respectively (p = .996). After drinking 150% of BM loss, NFB was also similar among conditions (p = .685). At the end of monitoring; however, significant differences were detected ($p < .001$): least squares means contrasts showed NFB estimated differences for W–SD = −0.25 kg (95% confidence interval—CI [−0.45, −0.06] kg, $p < .001$), W–SLM = −0.40 kg (95% CI [−0.60, −0.21] kg, $p < .001$), and SD–SLM = −0.15 kg (95% CI [−0.34, 0.04] kg, p = .126).

Fluid retention values for each condition were (mean, 95% CI): W = 39.6% [28.4%, 50.6%], SD = 55.6% [43.7%–67.5%], and SLM = (69.8% [61.5%–78.0%]). The other contrasts were not significant: SD–W = 16.1% (95% CI [−0.1%, 33.0%], $p = .066$) and SLM-SD = 14.2% (95% CI [−0.2%, 31.1%], p = .118).

Figure 2 compares diuresis over time for the three conditions. There was a significant interaction between time of measurement and condition ($p < .001$) for partial collection of urine. Specific contrast means are as follows (mean difference; 95% CI): immediately upon completing rehydration, W–SD = 74.2 ml; [−32.5, 180.9] ml (p = .172), W–SLM = 100.4 ml; [11.1, 189.7] ml (p = .028), SD–SLM = 26.2 ml; [−48.8, 101.1] ml (p = .491). After 1 hr, W–SD = 183.1 ml; [76.4, 289.7] ml ($p < .001$), W–SLM = 347.9 ml; [258.6, 437.2] ml ($p < .001$), SD–SLM = 164.8 ml; [89.9, 239.8] ml ($p < .001$). After 2 hr, W–SD = 46.2 ml; [−60.5, 152.9] ml (p = .393), W–SLM = 144.0 ml; [54.7, 233.3] ml (p = .002), SD–SLM = 97.8 ml; [22.9, 172.8] ml (p = .010). Finally, after 3 hr, W–SD = 17.9 ml; [−88.7, 124.6] ml (p = .740), W–SLM = 32.2 ml; [−57.1, 121.5] ml (p = .477), SD–SLM = 14.3 ml; [−60.7, 89.2] ml (p = .707).

Figure 3 shows accumulated urine volumes for all conditions. There is a significant interaction between time of measurement and condition ($p < .001$). Mean contrasts analysis showed (mean difference; 95% CI): immediately upon completing rehydration, W–SD = 74.1 ml; [−114.9, 263.1] ml (p = .440), W–SLM =100.3 ml; [−59.5, 260.1] ml (p = .217), SD–SLM = 26.2 ml; [−111.0, 163.4] ml (p = .707). After 1 hr, W–SD = 208.9 ml; [75.3, 342.6] ml (p = .002), W–SLM = 378.7 ml; [265.7, 491.7] ml ($p < .001$), SD–SLM = 169.8 ml; [72.8, 266.8] ml (p = .001). After 2 hr, W–SD = 298.7 ml; [165.1, 432.4] ml ($p < .001$), W–SLM = 576.8 ml; [463.9, 689.8] ml ($p < .001$), SD–SLM =278.1 ml; [181.1, 375.1] ml ($p < .001$). Finally, after 3 hr, W–SD =297.9 ml; [164.3, 431.6] ml ($p < .001$), W–SLM = 611.0 ml; [498.0, 724.0] ml ($p < .001$), SD–SLM = 313.1 ml; [216.1, 410.1] ml ($p < .001$).

GI Symptoms

GI symptoms were measured at preexercise, postexercise, and 60, 120, and 180 min, but beverages were ingested postexercise; only the results of 60, 120, and 180 min postexercise are shown. Most reported scores were low (0) or light to moderate (1–4) on the 0–9 scale. Accumulated scores were low: upper GI tract CS = 7 out of 63, lower tract CS = 5 out of 54, and systemic CS = 8 out of 36.

Table 1 Environmental and Physiological Variables by Condition

Variable	Water Mean (95% CI)	Sports drink Mean (95% CI)	Milk Mean (95% CI)	Sig. ($p < .05$)
USG (arrival)	1.012 [1.010, 1.015]	1.014 [1.012, 1.017]	1.014 [1.011, 1.017]	.581
Preexercise BM (kg)	66.5 [59.3, 73.6]	66.7 [59.5, 73.9]	66.5 [59.3, 73.7]	.998
Ingested volume (ml)	1,991.8 [1,765.2, 2,218.4]	1,998.8 [1,769.7, 2,227.8]	1,993.1 [1,765.8, 2,220.4]	.998
Dehydration (%)	−1.9 [−2.0, −1.9]	−2.0 [−2.0, −1.9]	−2.0 [−2.0, −1.9]	.715
Exercise time (min)	56.56 [52.03, 61.10]	56.56 [51.35, 61.78]	56.43 [50.79, 62.09]	.999
Temperature (°C)	32.14 [32.04, 32.23]	32.16 [32.05, 32.27]	32.21 [32.12, 32.31]	.529
Relative humidity (%)	71.29 [70.65, 71.94]	70.93 [70.32, 71.55]	70.98 [70.24, 71.72]	.708
Preexercise *z*PC GIS				
Upper	−0.30 [−0.41, −0.19]	−0.19 [−0.30, −0.08]	−0.34 [−0.45, −0.23]	.140
Lower	−0.13 [−0.35, −0.08]	−0.13 [−0.35, −0.08]	−0.26 [−0.48, −0.04]	.609
Systemic	−0.95 [−1.05, −0.86]	−0.90 [−1.00, −0.80]	−0.98 [−1.08, −0.88]	.501
Postexercise *z*PC GIS				
Upper	0.09 [−0.43, 0.62]	−0.15 [−0.68, 0.37]	0.28 [−0.24, 0.81]	.495
Lower	−0.26 [−0.57, 0.04]	−0.00 [−0.30, 0.30]	−0.26 [−0.57, 0.04]	.375
Systemic	−0.32 [−0.81, 0.16]	−0.11 [−0.60, 0.37]	0.04 [−0.44, 0.53]	.560

Note. Sig. = significance; 95% CI = lower and upper limits of the 95% confidence interval; BM = body mass; *z*PC GIS = standardized combined scores for gastrointestinal symptoms; Upper = upper gastrointestinal tract; Lower = lower gastrointestinal tract; Systemic = systemic symptoms; USG = urine specific gravity.

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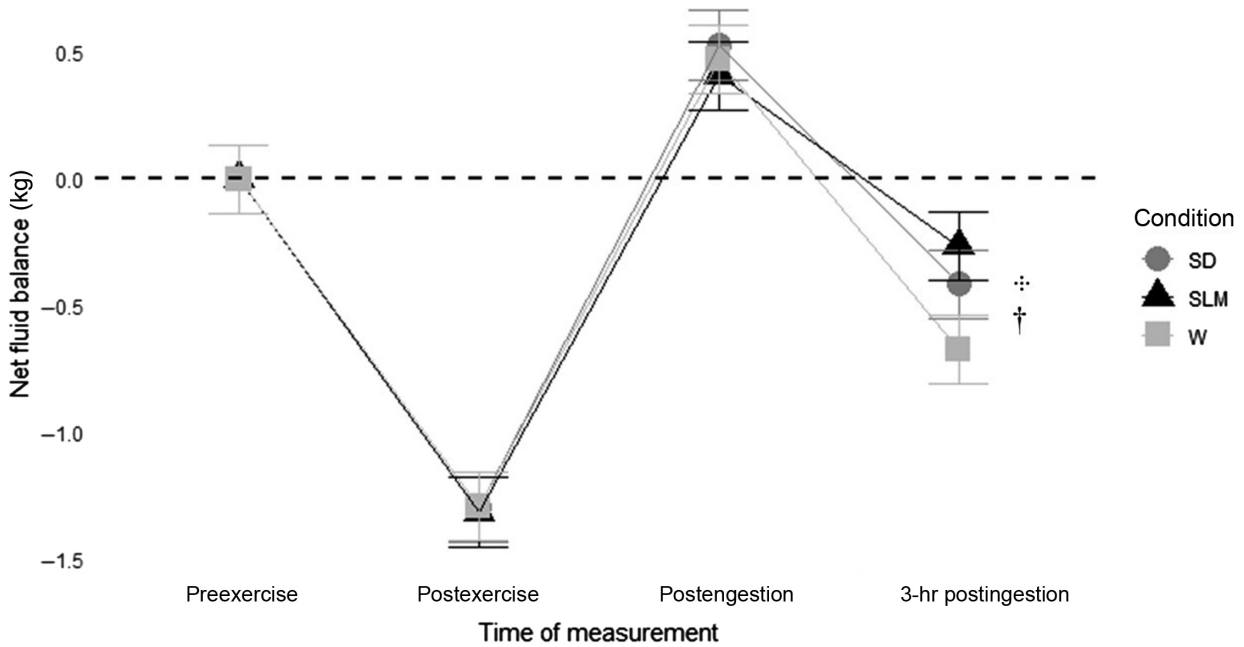

Figure 1 — Net fluid balance over time for SLM, SD, and W conditions ($n = 48$). Values are present as means (in kilograms) ± 95% CI. (✣) Significant difference between SLM and W; (†) Significant difference between W and SD. CI = confidence interval; SLM = skimmed, lactose-free milk; SD = sports drink; W = water.

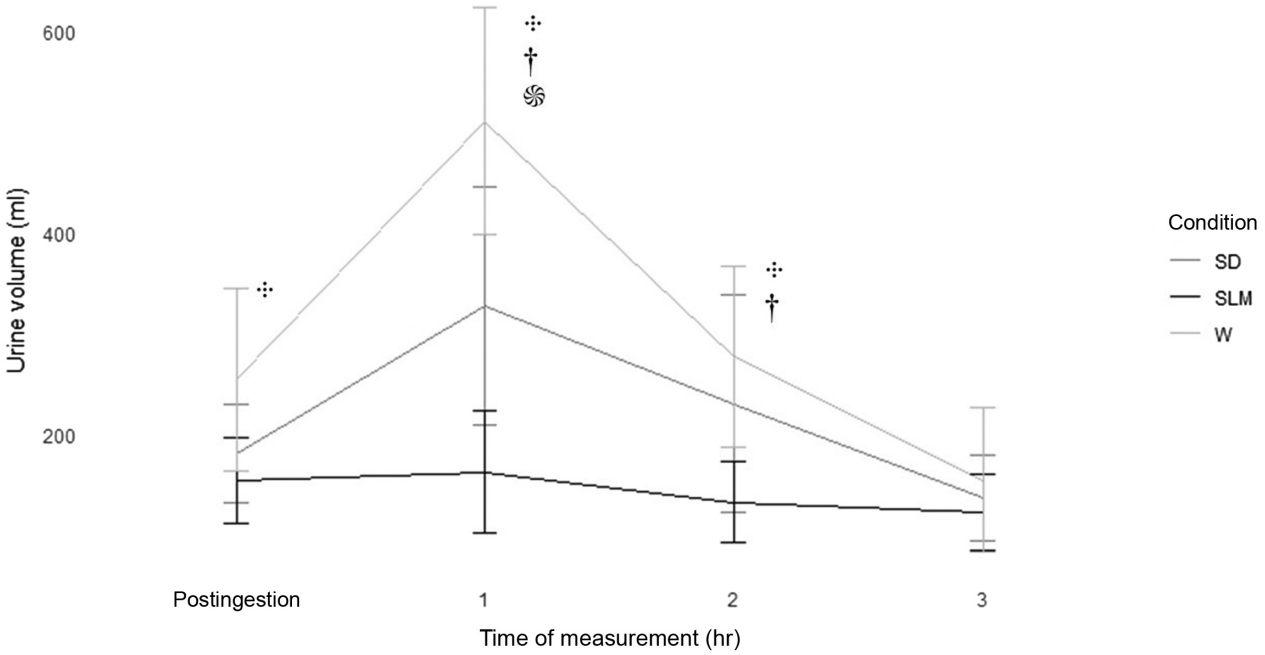

Figure 2 — Postingestion diuresis, 3-hr follow-up for SLM, SD, and W conditions ($n = 48$). Values are means (in milliliters) ± 95% CI. (✣) Significant difference ($p < .05$) between SLM and W. (†) Significant difference ($p < .05$) between SLM and SD. (✠) Significant difference ($p < .05$) between SD and W. CI = confidence interval; SLM = skimmed, lactose-free milk; SD = sports drink; W= water.

Table 2 shows CS for GI symptoms in each category (upper GI, lower GI, and systemic) and individual symptoms within each category. Data are from 60, 120, and 180 min postexercise for W, SD, and SLM, respectively. Median and range of scores are reported for each beverage and symptom/category. Overall, most symptoms had low median scores of 0–3, indicating mild or no GI issues across beverages at the times measured.

CS were standardized for analysis. The triple interaction among condition, time of measurement, and symptom category was not significant ($p = .904$; Figure 4). No significant double interactions were found between condition and time of measurement ($p = .458$), condition and symptom category ($p = .058$), or time of measurement and symptom ($p = .927$). Simple effects were significant: symptom ($p = .045$), condition ($p = .025$), and time of measurement ($p = .017$).

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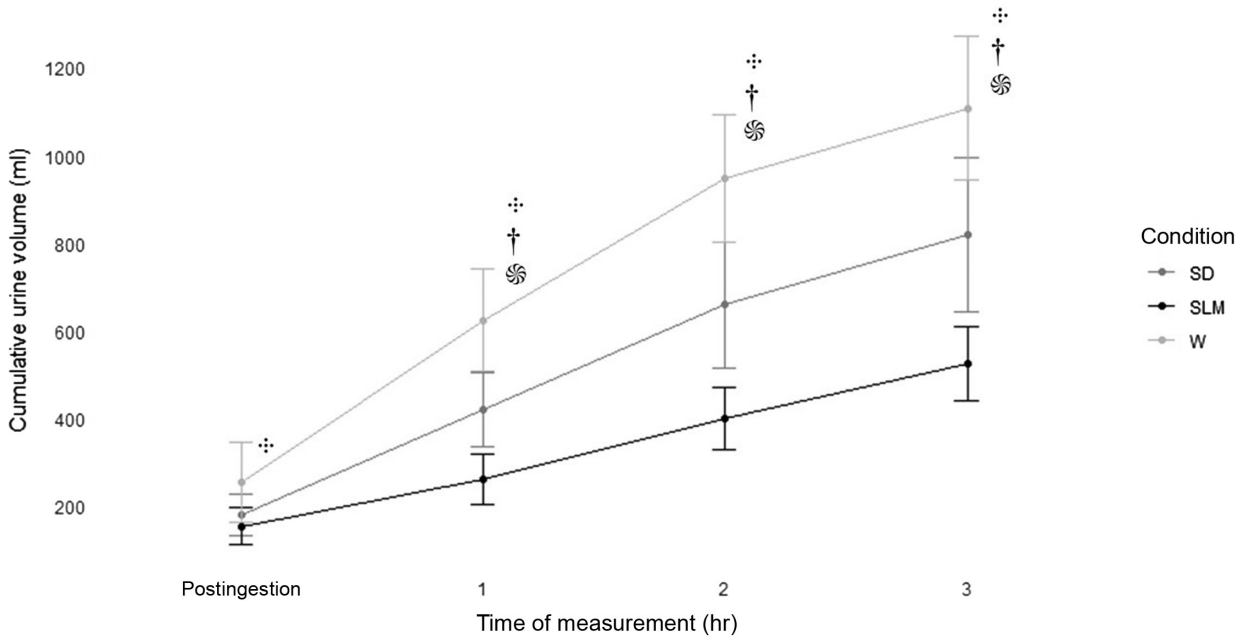

Figure 3 — Accumulated urine volume ($n = 48$) with a 3-hr follow-up after fluid ingestion (SLM, SD, and W). Mean (in milliliters) ± 95% CI. (✣) Significant difference ($p < .05$) between SLM and W. (†) Significant difference ($p < .05$) between SLM and SD. (✇) Significant difference ($p < .05$) between SD and W. CI = confidence interval; SLM = skimmed, lactose-free milk; SD = sports drink; W= water.

Table 2 Combined Raw Scores for Each Category (Upper GI Tract, Lower GI Tract, and Systemic) and for Individual GI Symptoms, by Categories 60, 120, and 180 min Postexercise

		Water	Sports drink	SLM
Sections	**Symptom**	***M* (range)**	***M* (range)**	***M* (range)**
Upper GI tract 63		0 (7)	0 (6)	0 (13)
	Reflux	0 (0)	0 (0)	0 (6)
	Heartburn	0 (0)	0 (3)	0 (3)
	Vomiting	0 (0)	0 (0)	0 (4)
	Nausea	0 (0)	0 (0)	0 (4)
	Thick saliva	0 (7)	0 (6)	0 (4)
	Belching	0 (1)	0 (3)	0 (4)
Lower GI tract 54		0 (8)	0 (3)	0 (5)
	Intestinal cramps	0 (1)	0 (0)	0 (0)
	Abdominal distension	0 (4)	0 (1)	0 (5)
	Abdominal pain	0 (1)	0 (0)	0 (2)
	Flatulence	0 (4)	0 (4)	0 (2)
	Urge to defecate	0 (1)	0 (0)	0 (2)
Systemic 36		3 (10)	3 (8)	3 (7)
	Muscle cramping	0 (7)	0 (1)	0 (3)
	Headache	1 (3)	0 (5)	0 (5)
	Urge to urinate	1 (8)	2 (8)	1 (7)
	Dizziness	0 (3)	0 (3)	0 (4)

Note. Cramping and loose stools/diarrhea have a value of 0 in both its median and range in all three conditions. GI = gastrointestinal; M = median; SLM = skimmed, lactose-free milk.

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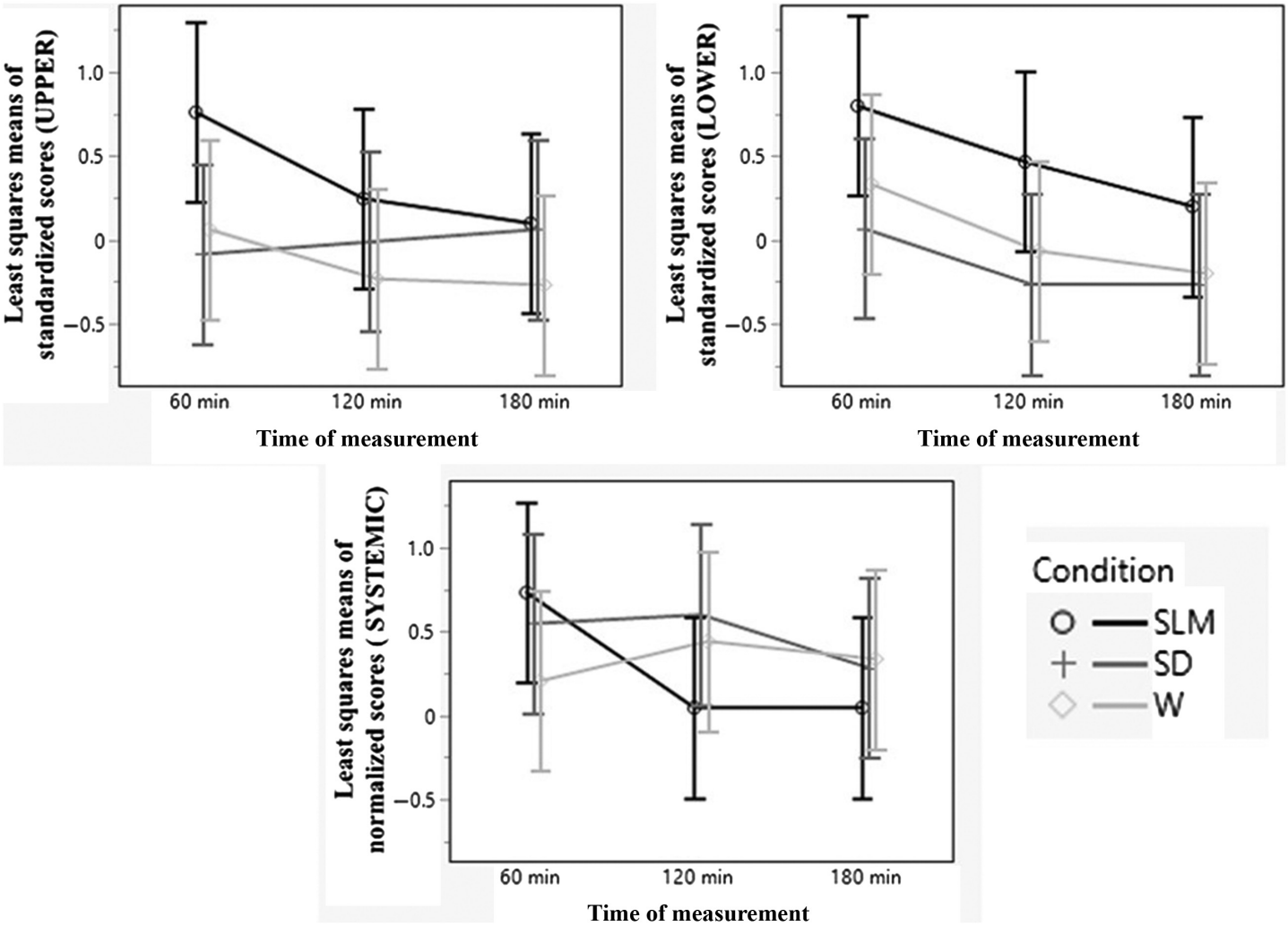

Figure 4 — Intestinal sensation least squares mean (standardized scores by sections) by condition and time of measurement. Values are means (standardized scores) ± 95% CI. CI = confidence interval; SLM = skimmed, lactose-free milk; SD = sports drink; W = water; UPPER = upper gastrointestinal tract; LOWER = lower gastrointestinal tract; SYSTEMIC = systemic symptoms.

Discussion

This study compared the rehydration effectiveness of W, SD, and SLM by measuring NFB, diuresis, fluid retention, and GI symptoms after ingesting ~2 L. Specifically, SLM showed higher NFB 3 hr postingestion, from greater fluid retention (~69%) versus W (~40%) although not different from SD (~56%); this supports SLM as an effective rehydration drink.

Volterman et al. (2014) reported greater retention for milk versus SD and W at 1 and 2 hr postexercise. James et al. (2011) found higher retention for milk (~55%) versus carbohydrate-electrolyte drink (~43%) at 4 hr ($p < .05$). Presently, SLM showed a higher NFB 3 hr postingestion (–0.26 kg) versus W (–0.67 kg) but not versus SD (–0.42 kg). Except for the latter nonsignificant difference, these results agree with previous studies (Berry et al., 2020; James et al., 2011; Shirreffs, Watson, et al., 2007) using different types of milk for postexercise rehydration. Nevertheless, in our study, SLM had lower TUV versus both W and SD.

SLM intake postexercise resulted in a lower diuresis versus W and SD. This aligns with Seery and Jakeman (2016) reporting lower diuresis 2, 3, 4, and 5 hr after milk intake versus W and SD postexercise in the heat (30 °C, 58% humidity). The effect could stem from milk's nutritional composition. Several studies (Baguley et al., 2016; Maughan et al., 2016; Campagnolo et al., 2017; Desbrow et al., 2014; Mallari et al., 2019) observed this with milk versus SDs, possibly due to slower gastric emptying from milk's higher energy content (≈1,415 kJ/L) versus the SD (≈1,000 kJ/L). Overall, the lower diuresis and higher retention with SLM suggest it has greater hydrating capacity versus SD or W postexercise.

Milk protein content could be an important mechanism for lower SLM diuresis. Milk proteins may stimulate antidiuretic hormone, and milk electrolytes could also increase plasma osmolality to activate antidiuretic hormone (Watson et al., 2008). Studies show dairy proteins' role in postexercise fluid retention and rehydration: Evans et al. (2018) found whey protein did not improve or inhibit retention versus maltodextrin. Hobson and James (2015) noted dairy proteins in drinks do not interfere with rehydration, James et al. (2013) suggested dairy proteins increase fluid retention, James et al. (2012) linked casein to greater retention from gastric coagulation, and Seifert et al. (2006) found higher retention with a protein and carbohydrate drink versus drinks with carbohydrate or plain W. Overall, these previous studies, along with the results of the present one, support a link between dairy protein and greater milk fluid retention.

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Nevertheless, from the present study it is not possible to reach conclusions regarding the specific ingredients responsible for the fluid retention qualities of SLM. The systematic, rigorous comparison of specific, commercially available drinks is useful to make practical recommendations to physically active people, but it is not compatible with determining the role of each component in fluid retention, as acknowledged by Shirreffs et al. (2007). In the present study, SLM and SD differed in protein, carbohydrates, fat, and potassium; new studies controlling one beverage ingredient at a time, particularly protein, are warranted.

Several studies (Aragón-Vargas, 2016; Maughan et al., 2004; Roy, 2008; Zhang et al., 2023) suggest skimmed milk may have slower gastric emptying versus W and SDs because of its higher energy and fat content. Our participants ingested ≈2 L of SLM, ~2,828 kJ. Participants in Clayton et al. (2014) drank ≈2.1 L, ≈3,696 kJ (a 10% carbohydrate solution); 40% of the fluid was still in the stomach after 2 hr. Possibly, some SLM may still be in the stomach after 3 hr in our case. Further research with methods like appearance of D_2O (deuterium) in the blood (Williams et al., 2023) is warranted to determine where the extra fluid from SLM resides in the body.

Regarding GI disturbances, the present study found no significant differences in their appearance or severity among conditions (W, SD, and SLM). GI symptom scores were low regardless of category, suggesting absent or mild problems on the 0–9 scale. This agrees with da Silva et al. (2021) who showed no severe symptoms after rehydration with a low-fat, lactose-free beverage with protein. Odell and Wallis (2021) pointed out even regular lactose-containing dairy beverages do not provoke GI disturbances in most lactose resistant people, after ingesting large volumes before, during, or after exercise.

In the present study, significant differences in GI symptoms were observed at preexercise time of measurement. These were higher in upper and lower GI tract, versus systemic symptoms, probably from the standardized breakfast consumed upon arrival. However, for all postexercise and postfluid ingestion time of measurements, no differences were identified among categories or time of measurement. The report of very mild symptoms after milk ingestion postexercise, and no significant differences among beverages, agrees with Karp et al. (2006), da Silva et al. (2021) and Russo et al. (2021), which show very low severity in reported GI symptoms.

Because of the nature of this study, it was not possible to blind the participants from the conditions, as the beverages were clearly distinguishable by taste. This may be considered a limitation, but given the fact that participants were not used to drinking milk after exercising, they would have been expected to rate their GI symptoms higher in that condition. This was not the case.

Conclusions

Together, the results from this study confirm that SLM is more effective than W and similar to a conventional SD for postexercise rehydration: Resulting diuresis is lower and therefore both fluid retention and NFB are higher than W, without causing greater GI tract disturbance.

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References

Adams, R. (1999). Revised physical activity readiness questionnaire. *Canadian Family Physician, 45*, Article 992.

Aragón-Vargas, L.F. (2016). Need of other elements. In F. Meyer, Z. Szygula, & B. Wilk (Eds.), *Fluid balance, hydration, and athletic performance* (p. 397). Taylor and Francis Group. https://doi.org/10.1201/b19037-25

Aragón-Vargas, L.F., Garzón-Mosquera, J.C., & Montoya-Arroyo, J. A. (2023). Voluntary hydration with skimmed lactose-free milk during exercise in the heat: Exploring effectiveness and tolerance. *Nutrients, 15*(9), Article 2069. https://doi.org/10.3390/nu15092069

Baguley, B.J., Zilujko, J., Leveritt, M.D., Desbrow, B., & Irwin, C. (2016). The effect of Ad libitum consumption of a milk-based liquid meal supplement vs. a traditional sports drink on fluid balance after exercise. *International Journal of Sport Nutrition and Exercise Metabolism, 26*(4), 347–355. https://doi.org/10.1123/ijsnem.2015-0229

Baker, L.B., & Jeukendrup, A.E. (2011). Optimal composition of fluid-replacement beverages. *Comprehensive Physiology, 4*(2), 575–620. https://doi.org/10.1002/cphy.c130014

Berry, C.W., Wolf, S.T., Murray, B., & Kenney, W.L. (2020). Hydration efficacy of a milk permeate-based oral hydration solution. *Nutrients, 12*(5), Article 1502. https://doi.org/10.3390/nu12051502

Campagnolo, N., Iudakhina, E., Irwin, C., Schubert, M., Cox, G.R., Leveritt, M., & Desbrow, B. (2017). Fluid, energy and nutrient recovery via ad libitum intake of different fluids and food. *Physiology & Behavior, 171*, 228–235. https://doi.org/10.1016/j.physbeh.2017.01.009

Clayton, D.J., Evans, G.H., & James, L.J. (2014). Effect of drink carbohydrate content on postexercise gastric emptying, rehydration, and the calculation of net fluid balance. *International Journal of Sport Nutrition and Exercise Metabolism, 24*(1), 79–89.

da Silva, C.D., de Oliveira, D.R., Perrone, Í.T., Fonseca, C.H., & Garcia, E.S. (2021). Low-fat, lactose-free and leucine-enriched chocolate cow milk prototype: A preliminary study on sensorial acceptability and gastrointestinal complaints following exhaustive exercise. *Journal of the International Society of Sports Nutrition, 18*(1), Article 14. https://doi.org/10.1186/s12970-020-00406-0

del Carmen Tocaa, M., Fernándezb, A., Orsic, M., Tabaccod, O., & Vinderolae, G. (2022). Lactose intolerance: Myths and facts. An

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update. *Archivos Argent de Pediatria, 120*(1), 59–66. https://doi.org/10.5546/aap.2022.eng.59

Desbrow, B., Jansen, S., Barrett, A., Leveritt, M.D., & Irwin, C. (2014). Comparing the rehydration potential of different milk-based drinks to a carbohydrate–electrolyte beverage. *Applied Physiology, Nutrition, and Metabolism, 39*(12), 1366–1372. https://doi.org/10.1139/apnm-2014-0174

Evans, G.H., Mattin, L., Ireland, I., Harrison, W., Yau, A.M., McIver, V., Pocock, T., Sheader, E., & James, L.J. (2018). Bolus ingestion of whey protein immediately post-exercise does not influence rehydration compared to energy-matched carbohydrate ingestion. *Nutrients, 10*(6), Article 769.

Garber, C.E., Blissmer, B., Deschenes, M.R., Franklin, B.A., Lamonte, M.J., Lee, I.M., Nieman, D.C., & Swain, D.P. (2011). American College of Sports Medicine position stand. Quantity and quality of exercise for developing and maintaining cardiorespiratory, musculoskeletal, and neuromotor fitness in apparently healthy adults: Guidance for prescribing exercise. *Medicine & Science in Sports & Exercise, 43*(7), 1334–1359. https://doi.org/10.1249/mss.0b013e318213fefb

Hobson, R., & James, L. (2015). The addition of whey protein to a carbohydrate–electrolyte drink does not influence post-exercise rehydration. *Journal of Sports Sciences, 33*(1), 77–84.

James, L.J., Clayton, D., & Evans, G.H. (2011). Effect of milk protein addition to a carbohydrate–electrolyte rehydration solution ingested after exercise in the heat. *British Journal of Nutrition, 105*(3), 393–399. https://doi.org/10.1017/S0007114510003545

James, L.J., Evans, G.H., Madin, J., Scott, D., Stepney, M., Harris, R., Stone, R., & Clayton, D.J. (2013). Effect of varying the concentrations of carbohydrate and milk protein in rehydration solutions ingested after exercise in the heat. *British Journal of Nutrition, 110*(7), 1285–1291.

James, L.J., Gingell, R., & Evans, G.H. (2012). Whey protein addition to a carbohydrate-electrolyte rehydration solution ingested after exercise in the heat. *Journal of Athletic Training, 47*(1), 61–66.

Jeukendrup, A., & Moseley, L. (2010). Multiple transportable carbohydrates enhance gastric emptying and fluid delivery. *Scandinavian Journal of Medicine & Science in Sports, 20*(1), 112–121. https://doi.org/10.1111/j.1600-0838.2008.00862.x

Karp, J.R., Johnston, J.D., Tecklenburg, S., Mickleborough, T.D., Fly, A.D., & Stager, J.M. (2006). Chocolate milk as a post-exercise recovery aid. *International Journal of Sport Nutrition and Exercise Metabolism, 16*(1), 78–91. https://doi.org/10.1123/ijsnem.16.1.78

Kurdak, S., Shirreffs, S., Maughan, R., Ozgünen, K., Zeren, C., Korkmaz, S., Yazici, Z., Ersöz, G., Binnet, M., & Dvorak, J. (2010). Hydration and sweating responses to hot-weather football competition. *Scandinavian Journal of Medicine & Science in Sports, 20*, 133–139.

Leiper, J.B., Maughan, R., & Murray, R. (2001). Gastric emptying and intestinal absorption of fluids, carbohydrates, and electrolytes. *Sports Drinks, 2000*, 89–128.

Mallari, M.F.T., Nana, A., Pinthong, M., Kongkum, S., & Chaunchaiyakul, R. (2019). Post-exercise ingestion of lactose-free skim milk affects thirst but not subsequent performance and net fluid balance of collegiate badminton athletes. *Malaysian Journal of Nutrition, 25*(3), Article 144. https://doi.org/10.31246/mjn-2018-0144

Maughan, R.J., Merson, S.J., Broad, N.P., & Shirreffs, S.M. (2004). Fluid and electrolyte intake and loss in elite soccer players during training. *International Journal of Sport Nutrition and Exercise Metabolism, 14*(3), 333–346. https://doi.org/10.1123/ijsnem.14.3.333

Maughan, R.J., Watson, P., Cordery, P.A., Walsh, N.P., Oliver, S.J., Dolci, A., Rodriguez-Sanchez, N., & Galloway, S.D. (2016). A randomized trial to assess the potential of different beverages to affect hydration status: Development of a beverage hydration index. *The American Journal of Clinical Nutrition, 103*(3), 717–723. https://doi.org/10.3945/ajcn.115.114769

Misselwitz, B., Butter, M., Verbeke, K., & Fox, M.R. (2019). Update on lactose malabsorption and intolerance: Pathogenesis, diagnosis and clinical management. *Gut, 68*(11), 2080–2091. https://doi.org/10.1136/gutjnl-2019-318404

Odell, O.J., & Wallis, G.A. (2021). The application of lactose in sports nutrition. *International Dairy Journal, 116*, Article 104970. https://doi.org/10.1016/j.idairyj.2020.104970

Pérez-Castillo, Í.M., Williams, J.A., López-Chicharro, J., Mihic, N., Rueda, R., Bouzamondo, H., & Horswill, C.A. (2023). Compositional aspects of beverages designed to promote hydration before, during, and after exercise: Concepts revisited. *Nutrients, 16*(1), Article 17.

Pfeiffer, B., Stellingwerff, T., Hodgson, A.B., Randell, R., Pöttgen, K., Res, P., & Jeukendrup, A.E. (2012). Nutritional intake and gastrointestinal problems during competitive endurance events. *Medicine & Science in Sports & Exercise, 44*(2), 344–351. https://doi.org/10.1249/MSS.0b013e31822dc809

Roy, B.D. (2008). Milk: The new sports drink? A Review. *Journal of the International Society of Sports Nutrition, 5*(1), Article 15. https://doi.org/10.1186/1550-2783-5-15

Russo, I., Della Gatta, P.A., Garnham, A., Porter, J., Burke, L.M., & Costa, R.J. (2021). Does the nutritional composition of dairy milk based recovery beverages influence post-exercise gastrointestinal and immune status, and subsequent markers of recovery optimisation in response to high intensity interval exercise? *Frontiers in Nutrition, 7*, Article 622270. https://doi.org/10.3389/fnut.2020.622270

Sawka, M., Burke, L., Eichner, E., Maughan, R., Mountain, S., & Stachenfeld, N. (2007). Exercise and fluid replacement: Position Stand American College of Sports Medicine. *Medicine & Science in Sports & Exercise, 39*(2), 377–390. https://doi.org/10.1249/mss.0b013e31802ca597

Seery, S., & Jakeman, P. (2016). A metered intake of milk following exercise and thermal dehydration restores whole-body net fluid balance better than a carbohydrate–electrolyte solution or water in healthy young men. *British Journal of Nutrition, 116*(6), 1013–1021. https://doi.org/10.1017/S0007114516002907

Seifert, J., Harmon, J., & DeClercq, P. (2006). Protein added to a sports drink improves fluid retention. *International Journal of Sport Nutrition and Exercise Metabolism, 16*(4), 420–429. https://doi.org/10.1123/ijsnem.16.4.420

Shirreffs, S.M., Aragon-Vargas, L.F., Keil, M., Love, T.D., & Phillips, S. (2007). Rehydration after exercise in the heat: A comparison of 4 commonly used drinks. *International Journal of Sport Nutrition and Exercise Metabolism, 17*(3), 244–258. https://doi.org/10.1123/ijsnem.17.3.244

Shirreffs, S.M., Watson, P., & Maughan, R.J. (2007). Milk as an effective post-exercise rehydration drink. *British Journal of Nutrition, 98*(1), 173–180. https://doi.org/10.1017/S0007114507695543

Volterman, K.A., Obeid, J., Wilk, B., & Timmons, B.W. (2014). Effect of milk consumption on rehydration in youth following exercise in the heat. *Applied Physiology, Nutrition, and Metabolism, 39*(11), 1257–1264. https://doi.org/10.1139/apnm-2014-0047

Watson, P., Love, T.D., Maughan, R.J., & Shirreffs, S.M. (2008). A comparison of the effects of milk and a carbohydrate-electrolyte drink on the restoration of fluid balance and exercise capacity in a hot, humid environment. *European Journal of Applied Physiology, 104*(4), 633–642. https://doi.org/10.1007/s00421-008-0809-4

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Williams, J.A., Choe, Y.S., Skoufos, L., Huang, Y., Zhong, P.S., & Horswill, C.A. (2023). Hydration dynamics: A model to study absorption, plasma volume expansion, and fluid retention: 822. *Medicine & Science in Sports & Exercise, 55*(Suppl. 9), Article 287. https://doi.org/10.1249/01.mss.0000982412.90733.1f

Zhang, P., Iqbal, S., Deng, R., Duan, X., Han, H., Chen, X.D., & Wu, P. (2023). Impact of elderly gastrointestinal alterations on gastric emptying and enzymatic hydrolysis of skim milk: An in vitro study using a dynamic stomach system. *Food Chemistry, 402*, Article 134365. https://doi.org/10.1016/j.foodchem.2022.134365

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